

all major communication standards so that it is convenient for users.

What more has come up since then?

Newer standards and modular upgrade options. The signal generator market has not witnessed revolutionary changes in the past few years. Hence, the release of new models or upgraded versions of particular instruments depend mainly on the changes brought about in testing standards.

Other changes seen in new models are widened specification range and enhancement of existing features. With the introduction of modular platforms, the upgradation of an instrument has become much easier for both manufacturers and users.

FPGA onboard. There are applications where data is required to be processed at a very high rate without any lag or loss of information. Field programmable gate array (FPGA) is a hardware based processor that processes incoming data on a point-to-point basis, unlike software based processing that requires buffering. This means, the user gets immediate response and high processing frequency as there is no buffer time and overhead time involved. Such FPGA processors are used in many modular platforms available these days.

Increased demand for specialised signal generators

On speaking to the spokespersons for this story, it was felt that the niche market for signal generators is going somewhat steady.

Education. Every electronics lab in an educational institution has at least one simple, low-frequency function generator for performing basic electronics experiments. With a large number of engineering colleges being set up every year, academia is one area where signal generators are being sold in good numbers. These usually generate the basic sine, square and triangle

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lar waves at a frequency that ranges from a few hertz to a few gigahertz. The user interface (UI) for these academia-focused instruments is simple to understand and operate.

Defence R&D. The purchase of high-frequency signal generators that can create custom signals generally comes from defence and aerospace development laboratories such as Defence Research & Development Organisation (DRDO). They also use pulse-signal generators for radar application.

Semiconductor. Many semiconductor firms such as Freescale, Intel, Qualcomm and Texas Instruments perform their chip design validation in India. Whether to check logic high and logic low of a chip, or to analyse the equivalent analogue output for a digital input, a signal generator is important.

Purchase of signal generators is notable in telecom (due to the continuously-evolving communication standards), light emitting diode (LED) lights and drivers manufacturing, and electronic device repair shops, among others.

Influenced by indigenous manufacturing

Signal generators are instruments that have not evolved much over the years, except for the introduction of modular platforms. Though customers working on advanced applications

TABLE II
Various Manufacturers and Distributors

Manufacturer	Partner/Distributor
Anritsu	Anritsu India Pvt Ltd Meera Agencies Pvt Ltd Optimized Solutions Ltd Peridot Technologies Scientech Technologies Pvt Ltd Sinetec Automation
Keysight	Keysight Technologies India Pvt Ltd
OWON-MetroQ	Metro Electronic Products
National Instruments	Captronic Systems Pvt Ltd Digilogic Systems Pvt Ltd National Instruments India NexGEN Consultancy Pvt Ltd Optimized Solutions Pvt Ltd
Tektronix	Aarjay International Pvt Ltd Convergent Technologies Cyronics Instruments Pvt Ltd Optimized Solutions Pvt Ltd Peridot Technologies Primetech Instruments Pvt Ltd RS Components & Controls (India) Ltd SPI Engineers Pvt Ltd Techno Scientific Co. Tektronix (India) Pvt Ltd Vishal Vyapar Vikash Vitronics (India)

opt for a broader range of parameters or integrated instruments, basic models of signal generators are still in demand.

With the launch of 'Make in India' campaign and push by central and state governments for indigenous manufacturing, manufacturers and distributors are hoping that five years down the line, the signal generator market would see significant growth in revenue. ●